

PROJECT SUTRA

Ultimate System Subsystem Architecture & First-Principles Roadmap (v1)

🎯 Ultimate Target Vision Included

🌱 First Principles Pacing

🔗 Clickable Learning Links

✅ Team Progress Checklist

AUTONOMOUS DRONE SWARM MISSION: Forest & Ruins Search & Rescue

The art illustration delivered visual encyclopedia in autonomous drone swarm mission in action. Lamsagan he hights and won autonomous drone can urients the funnital and swarm storm intpurses one of throwi-dat to contact a hiendly encorment.

🕒 Choices are highly autonomous drones of naiving creations, scams, nays and auto-matid data comaining.

- 1 Thermal Heat Imaging;
Detecting fire/human activity
- 2 Microwave Radar;
Building/Topo mapping
- 3 RGB Camera;
Real-time video

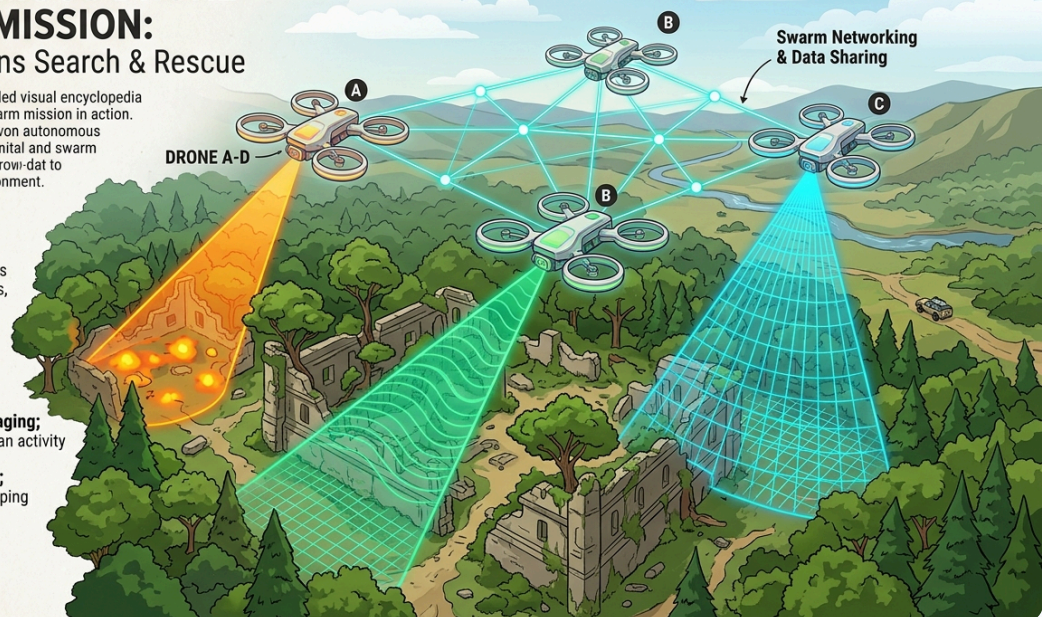


Figure 1.1: SUTRA Autonomous Multi-Drone Swarm Operation in Disaster & GPS-Denied Zones

1

The Ultimate SUTRA Subsystem Architecture & Target Vision

To ensure all five team members understand the **ultimate technical destination** they are building toward, this section details the complete target architecture of each subsystem: Autonomous Flight/GNC, Swarm Communication Mesh, Tri-Modal AI Perception, Ground Control Station (GCS), and Gazebo Hardware Abstraction (HAL).

HOW DRONE SWARMS WORK: A 4-LAYER SYSTEM

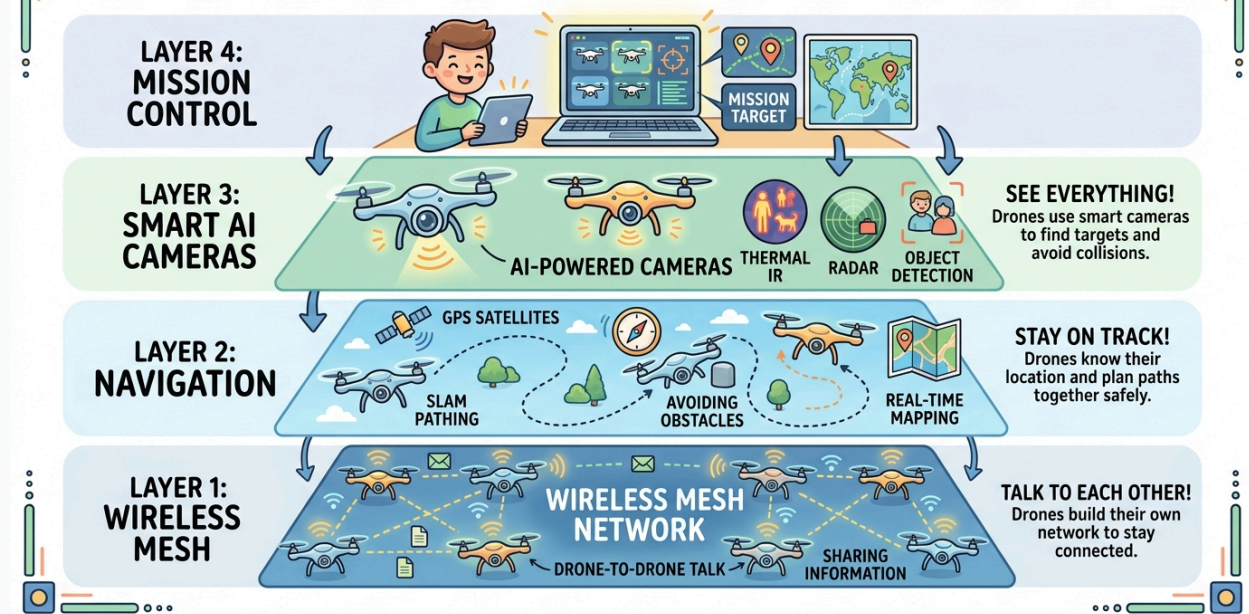


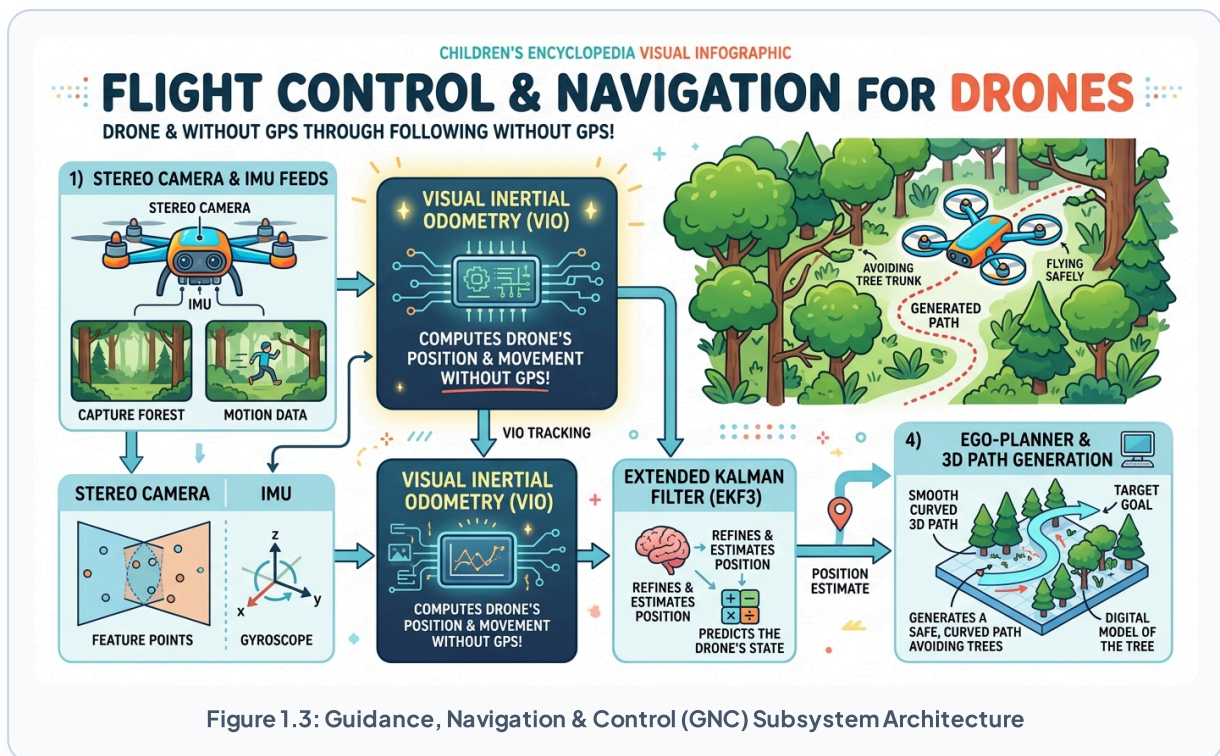
Figure 1.2: SUTRA 4-Layer System Architecture

Subsystem A: Autonomous Navigation, Flight Control & GNC

Lead Engineer: Rohith Kumar

Ultimate Target Vision: An autonomous GPS-denied Guidance, Navigation & Control stack capable of 3D trajectory generation, voxel-grid obstacle avoidance, and visual-inertial odometry.

- **Flight Controller Interface:** PX4 Autopilot running Offboard Control mode via MicroXRCE-DDS / MAVROS 2 bridge.
- **State Estimation & GPS-Denied Fallback:** Visual-Inertial Odometry (VIO) using OpenVINS / VINS-Mono paired with down-facing optical flow for position hold when GPS drops.
- **3D Voxel Mapping:** Real-time OctoMap 3D occupancy grid generation using stereo depth / LiDAR rangefinder inputs.
- **Multi-Agent Swarm Collision Avoidance:** Optimal Reciprocal Collision Avoidance (ORCA) algorithm preventing mid-air collisions.





Subsystem B: Swarm Communication Mesh & AI Neural Encoders

Lead Engineer:
Nikhil

Ultimate Target Vision: A resilient, self-healing ad-hoc mesh network utilizing Deep Joint Source-Channel Coding (Deep JSCC) to transmit target alerts under severe RF interference.

- **Ad-Hoc Network Protocols:** Hybrid 802.11s Wi-Fi mesh + long-range 868MHz LoRa telemetry for emergency heartbeats.
- **Deep JSCC Neural Coding:** Encoder/decoder neural networks that compress imagery directly into robust channel symbols under low SNR conditions.
- **Dynamic Consensus & Failover:** RAFT-based leader election protocol allowing seamless re-election when the swarm leader disconnects.
- **Bandwidth Packet Thinning:** Priority queue system dropping non-critical diagnostic packets when packet loss exceeds 50%.

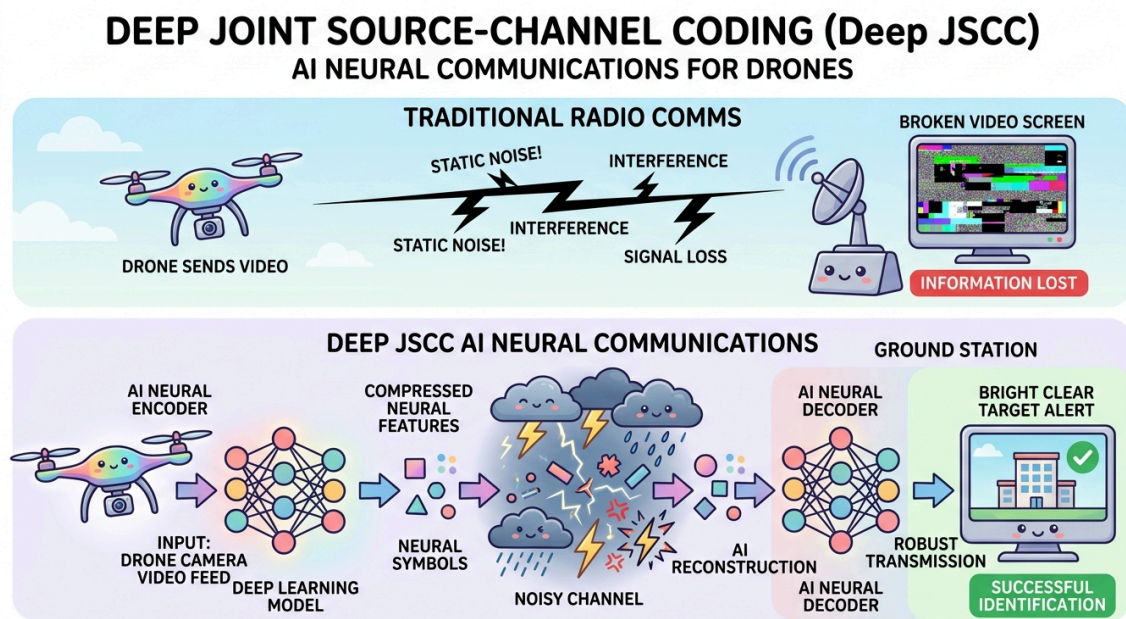


Figure 1.4: Deep JSCC Swarm Communication & Neural Channel Encoder Architecture

Subsystem C: Tri-Modal AI Perception & Sensor Fusion

Lead Engineer: Vedanth Sai Ram

Ultimate Target Vision: A multi-spectral vision pipeline fusing RGB, Long-Wave Infrared (LWIR) Thermal, and 77GHz mmWave FMCW Radar for target detection in smoke, fog, and zero-light.

- **Object Detection Engine:** YOLOv8–Nano neural network optimized with TensorRT / ONNX for 30+ FPS edge inference.
- **Hardware Target Acceleration:** On-device NPU processing using NetraSemi / Jetson Orin edge AI chips.
- **Tri-Modal Sensor Fusion:** Spatial cross-attention alignment merging visual bounding boxes, thermal heat maps, and micro-Doppler radar signatures.
- **Target Geolocation:** Raycasting bounding box centroids onto 3D elevation maps to output precise GPS target coordinates.

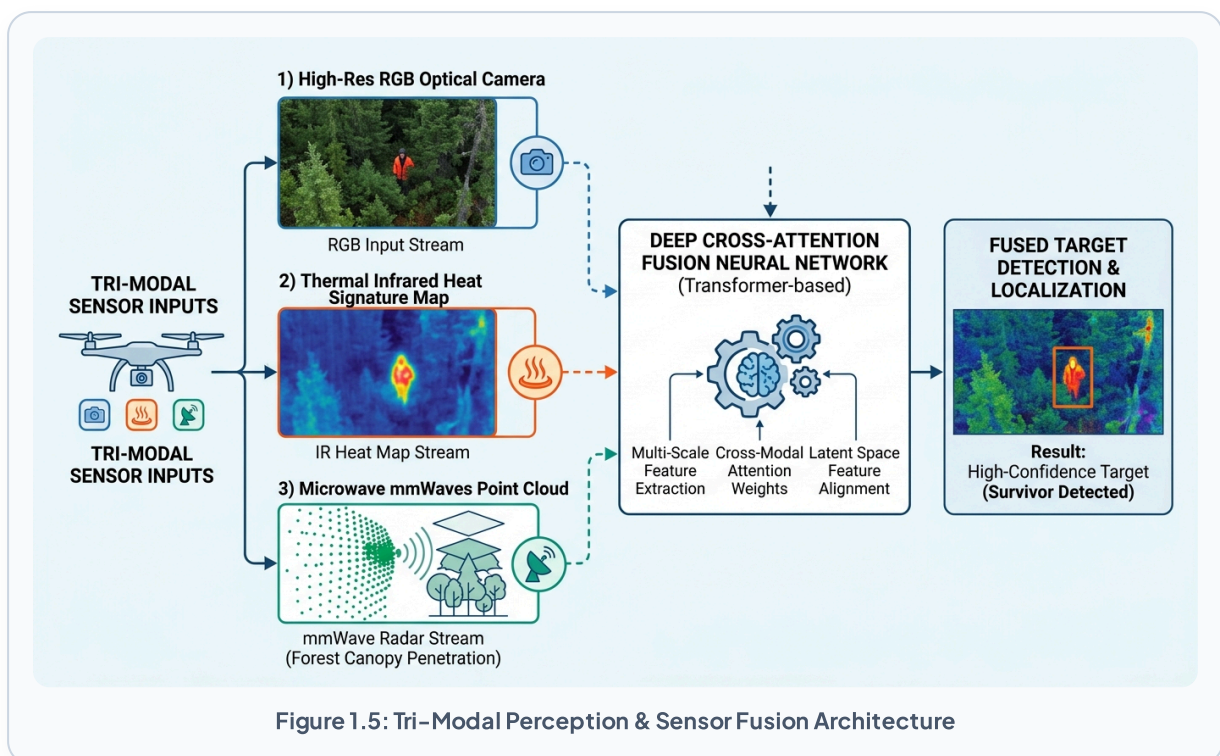


Figure 1.5: Tri-Modal Perception & Sensor Fusion Architecture

Subsystem D: 3D GIS Ground Control Station & HSI Dashboard

Lead Engineer: Siva
Kesava

Ultimate Target Vision: A modern web-based 3D Human-System Interface (HSI) rendering multi-drone telemetry, target alerts, and dual-redundant emergency overrides.

- **3D Satellite Map Engine:** React.js + Mapbox GL JS rendering interactive 3D terrain elevation and real-time drone markers.
- **Telemetry HUD Widgets:** WebGPU-powered artificial horizon, battery levels, link status, and thermal stream video feeds.
- **Operator Emergency Controls:** One-click Return-to-Launch (RTL), geo-fence containment boundaries, and manual swarm land triggers.
- **Data Caching:** Local SQLite / IndexedDB buffer preserving telemetry trails during network blackouts.

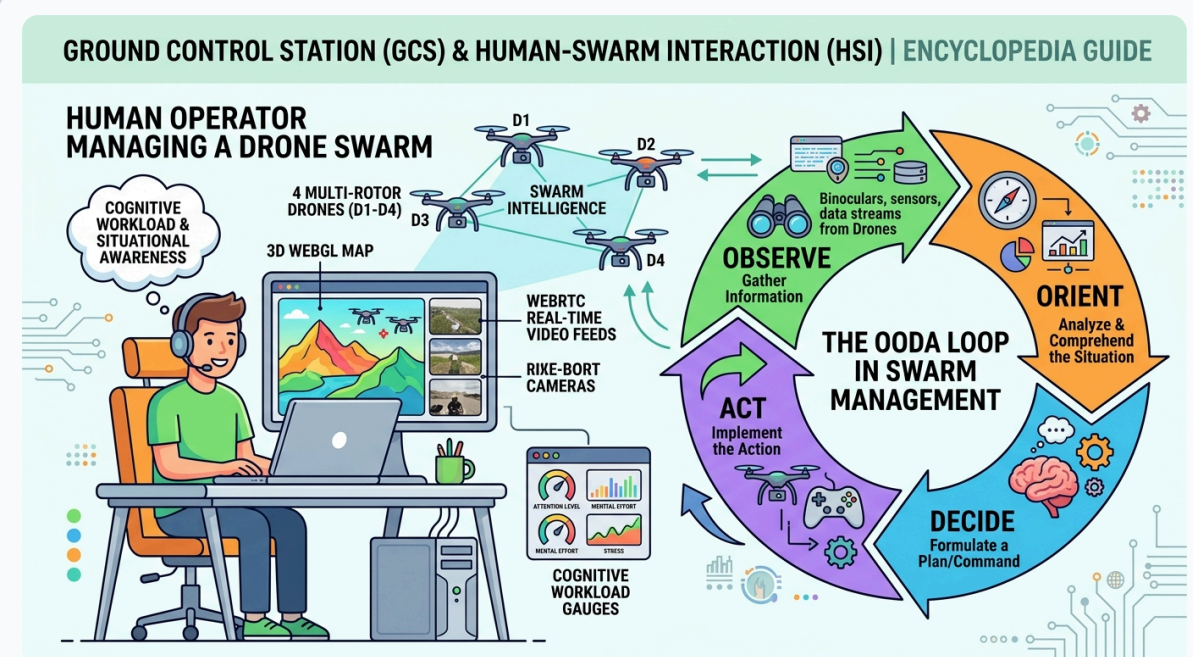


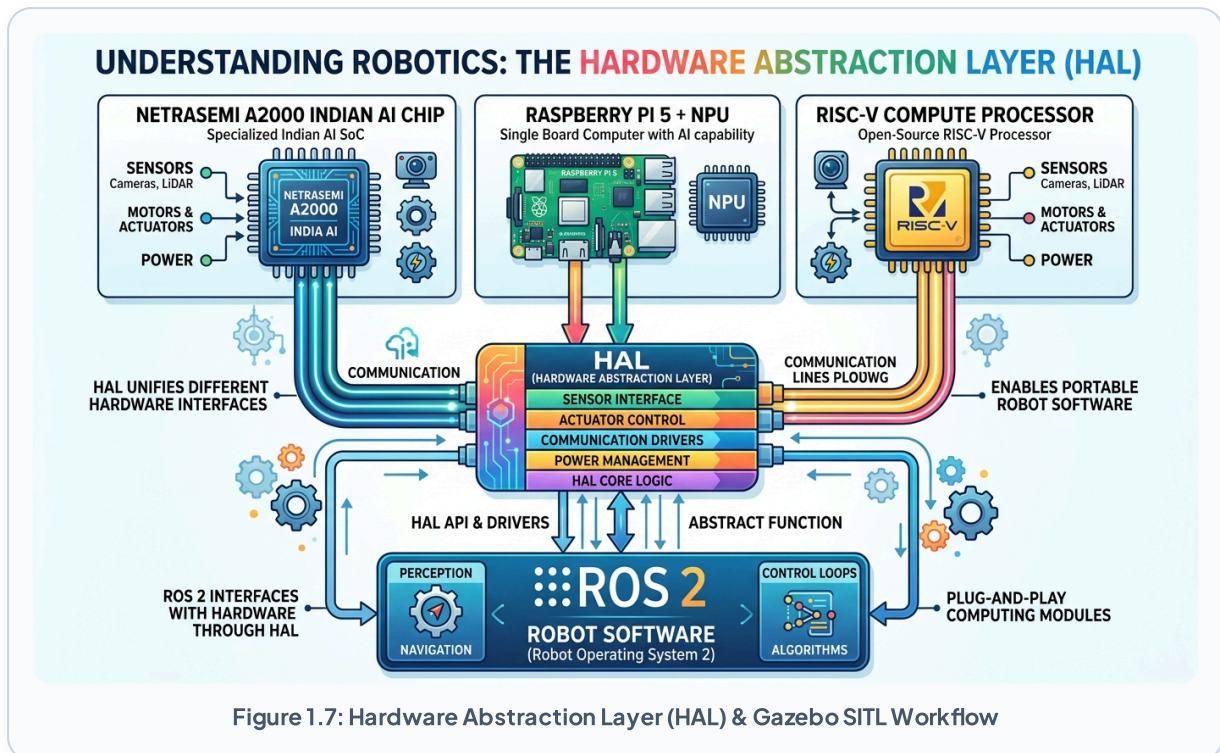
Figure 1.6: 3D GIS Ground Control Station (GCS) Dashboard Interface

⚙️ Subsystem E: Hardware Abstraction Layer (HAL) & Gazebo SITL

Lead Engineer:
Harika

Ultimate Target Vision: A high-fidelity Gazebo Harmonic 3D simulation environment enabling end-to-end multi-vehicle Software-in-the-Loop (SITL) mission testing.

- **Photorealistic 3D SDF Worlds:** Gazebo Harmonic simulation worlds modeling forest rubble, elevation changes, and atmospheric fog.
- **Sensor Plugins:** Simulated RGB cameras, thermal camera models, 77GHz mmWave radar plugins, and IMU/Barometer noise models.
- **Automated Launch Orchestrator:** ROS 2 launch scripts spinning up multi-vehicle PX4 SITL instances with independent namespaces (`/drone1`, `/drone2`).



2 The First-Principles 12-Week Master Learning Strategy

With the **ultimate technical goal** clearly defined above, Team Offgrid follows a **12-Week 3-Stage Progressive Roadmap** starting from first principles.



Figure 2.1: The 12-Week 3-Stage Progressive Engineering Roadmap

Stage 1: Shared Team Foundations (Weeks 1–4)

All 5 Engineers Together

Before role splitting, all five team members (Rohith, Nikhil, Vedanth, Siva, and Harika) master core tools and mathematical intuition together.

17 **Week 1: Linux CLI, Shell Scripting, Git & Workflows**

- Terminal navigation (`ls`, `cd`, `grep`, `chmod`, `systemctl`), environment variables, Bash scripting.
- Git version control (`branch`, `merge`, `rebase`, `PRs`), GitHub issue tracking, Markdown documentation.

 Linux Journey (Free)

 Automate the Boring Stuff (Al Sweigart)

 Git Immersion

17 **Week 2: Python Data Structures, OOP & Virtual Envs**

- Python object-oriented programming (Classes, Inheritance, Dictionaries, List comprehensions).
- Environment management (`venv`, `pip`), JSON handling, file I/O, error handling.

 FreeCodeCamp Python for Everybody

 RealPython OOP Tutorial

17 **Week 3: Essential Math for Autonomous Systems**

- Vectors, matrices, dot/cross products, matrix multiplication, eigenvalues.
- 2D/3D coordinate transformations, roll/pitch/yaw angles, rotation matrices, quaternion intuition.
- Basic probability & Gaussian distribution (foundations for Kalman Filters).

 3Blue1Brown: Essence of Linear Algebra

 Khan Academy Linear Algebra

17 **Week 4: Scientific Libraries & Socket Networking**

- **NumPy**: Vectorized math operations, matrix inversion, array slicing.
- **Matplotlib**: 2D plotting, trajectory plotting, real-time animation.
- **OpenCV**: Image reading, color space conversions (RGB → HSV), thresholding, contour drawing.
- **Networking**: Python `socket` module (UDP vs TCP sockets, packet encoding/decoding).

 SciPy NumPy Quickstart

 OpenCV Python Tutorial (Murtaza)

 Python Socket Programming (FreeCodeCamp)

Stage 2: Mini Systems & Foundation Capstones (Weeks 5–8)

Pair & Individual
Projects

Build small, working sub-systems from scratch to prove concepts before touching complex robotics frameworks.

Capstone 1 (Week 5): Socket Telemetry Relay

Owner: Nikhil & Rohith

Write a multithreaded Python UDP server/client that transmits simulated telemetry JSON(`{"drone_id": "D1", "lat": 17.38, "battery": 95}`) across terminals at 10Hz.

Capstone 2 (Week 6): OpenCV Target Tracker

Owner: Vedanth

Build a Python script that opens a camera feed, detects a specific color marker (e.g. orange cone), draws bounding boxes, and calculates centroid offsets.

Capstone 3 (Week 7): Web 3D Map Dashboard

Owner: Siva Kesava

Create a responsive HTML/React web page using Mapbox GL JS that receives mock drone telemetry over WebSocket/HTTP and updates 3D map pins live.

Capstone 4 (Week 8): 2D Robot Kinematics Sim

Owner: Harika

Build a 2D differential drive robot simulator in Python/Pygame that moves toward goal waypoints while steering around static circular obstacles.



Stage 3: Specialized SUTRA Role Onboarding (Weeks 9–12)

SUTRA Drone Stack
Integration

With foundations and mini-capstones complete, the team branches into specialized SUTRA modules and integrates Gazebo SITL simulation.

July 17

Week 9: ROS 2 Jazzy Architecture & Workspace Setup

ROS 2 workspace setup, nodes, publishers, subscribers, custom `.msg` / `.srv` definitions, QoS profiles.`



Official ROS 2 Jazzy Docs



Articulated Robotics ROS 2

July 17

Week 10: Gazebo SITL & Single Vehicle Control

Installing Gazebo Harmonic (``gz sim``), PX4 SITL integration, offboard flight control via MicroXRCE-DDS / MAVROS bridge.



Gazebo Official Docs



Official PX4 ROS 2 Guide

July 17

Week 11: Multi-Drone Swarm & Sensor Integration

Swarm ROS 2 namespaces (``/drone1` , `/drone2``), leader heartbeat consensus, multi-spectral camera plugin integration.

July 17

Week 12: Baseline System Qualification & Gate G1–G6 Audits

Execute full end-to-end mission qualification in Gazebo SITL across all 6 Verification Gates.

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Interactive & Printable Team Progress Checklist

Use this interactive checklist to mark off skills, tools, math concepts, and capstone milestones as you master them!

Stage 1 Checklist: Shared Foundations (Weeks 1–4)

☐ Linux CLI (`ls`, `cd`, `grep`, `chmod`)

☐ Git Branching, Merging & PRs

☐ Python OOP (Classes & Inheritance)

☐ Python `venv` & `pip` Package Manager

☐ Linear Algebra (Vectors & Matrices)

☐ 3D Math (Roll/Pitch/Yaw & Quaternions)

☐ NumPy Vectorized Operations

☐ Matplotlib Trajectory Plotting

☐ OpenCV Color Thresholding & Contours

☐ Python UDP/TCP Socket Networking

Stage 2 Checklist: Mini System Capstones (Weeks 5–8)

☐ Capstone 1: Telemetry UDP Socket Relay

☐ Capstone 2: OpenCV Target Tracker

☐ Capstone 3: Web 3D Mapbox Dashboard

☐ Capstone 4: 2D Kinematics Simulator

Stage 3 Checklist: Specialized SUTRA Modules (Weeks 9–12)

☐ **Rohith** : PX4 SITL 2m Takeoff Script

☐ **Rohith** : OctoMap 3D Obstacle Avoidance

☐ **Nikhil** : UDP JSON Swarm Heartbeat Mesh

☐ **Nikhil** : Priority Alert Packet Thinning

☐ **Vedanth** : YOLOv8–Nano Target Detection

☐ **Vedanth** : Dual RGB+Thermal BBox Alignment

☐ **Siva** : 3D Mapbox Satellite Pin Render

☐ **Siva** : GCS Emergency RTL Overrides

☐ **Harika** : Gazebo Harmonic 3D World Launch

☐ **Harika** : Gate G1–G6 SITL Verification

4 R&D Risk Audit & Tiered Subsystem Matrix

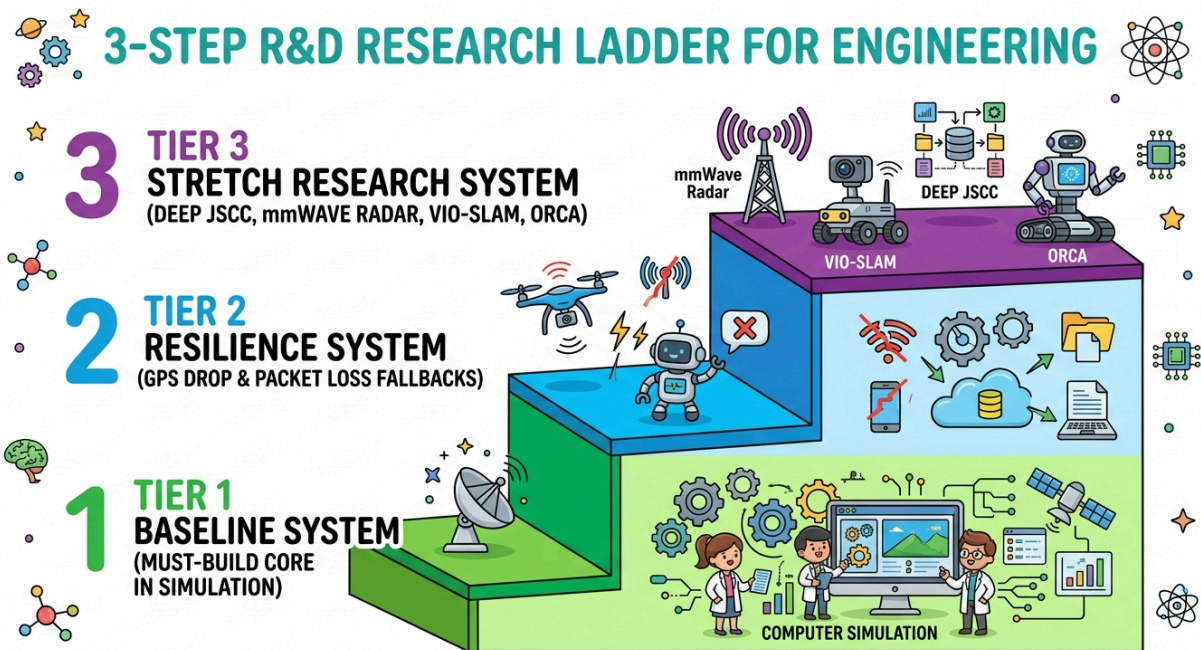


Figure 4.1: R&D Tiered Research Ladder (Tier1 Baseline → Tier 2 Resilience → Tier 3 Stretch)

Subsystem Module	Core Must-Build (Tier 1)	Resilience Fallback (Tier 2)	Stretch Research Path (Tier 3)
Navigation & GNC	Waypoint Navigation & OctoMap 3D obstacle avoidance	GPS drop fallback to VIO / Optical Flow pose hold	VINS-Mono / OpenVINS + ORCA Swarm Avoidance
Swarm Mesh Comms	JSON Alert Packets over UDP & 868MHz LoRa link	Priority alert packet thinning under loss	Deep JSCC Neural Channel Autoencoders
AI Perception	YOLOv8-Nano RGB + Thermal IR Late Fusion	Fallback to single thermal channel in zero-light	77GHz mmWave FMCW Radar Cross-Attention Fusion
GCS Dashboard	Live 3D Mapbox Map, Battery/HUD, RTL Overrides	Local SQLite telemetry caching during dropouts	Operator Cognitive Workload Manager & WebGPU

R&D SYSTEM DEPENDENCY GRAPH FOR ROBOTICS: DEVELOPMENT PIPELINE

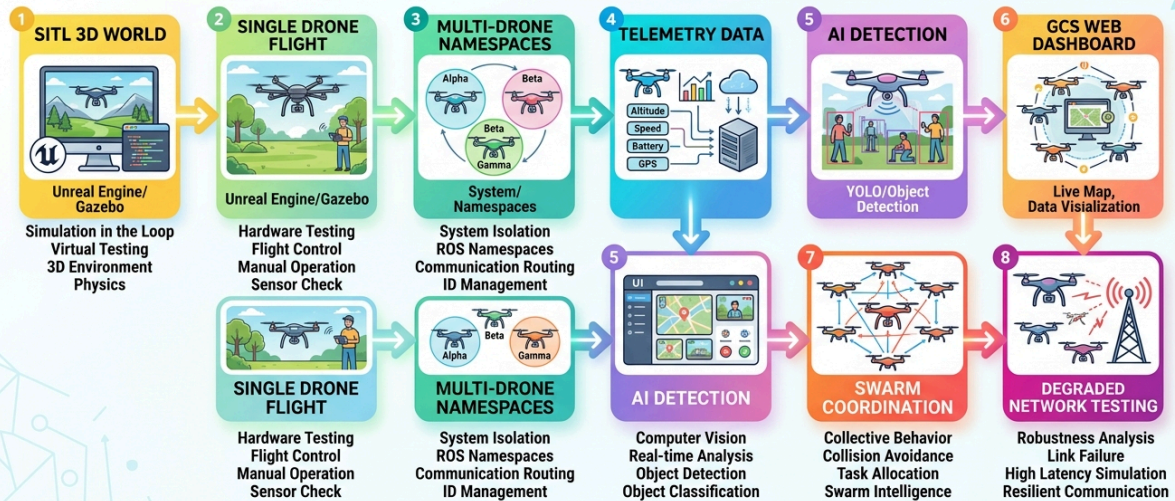


Figure 4.2: R&D System Dependency Graph

5 Quantified & Instrumented Verification Gates (G1–G6)

6 SIMULATION TESTING GATES FOR DRONE SWARMS



Figure 5.1: The 6 Quantified SITL Simulation Testing Gates (G1 to G6)

Gate ID	Milestone Objective	Quantified Success Criteria	Failure & Timeout Threshold
Gate G1	Single Drone Autonomous Flight	Takeoff, 10m square waypoint circuit, and auto-landing.	Max lateral error > 1.0 m; Altitude drift > 0.2 m.
Gate G2	Swarm Leader Failover	Primary leader killed; remaining peers re-elect leader.	Logical election > 500 ms; network consensus timeout > 2.0 s.
Gate G3	AI Target Identification	Spot human model in simulated forest rubble.	Precision < 85%, Recall < 80%, Geo-error > 1.5 m.
Gate G4	Degraded Network Comms	Inject 60% RF packet loss across Wi-Fi mesh.	Alert delivery ratio < 90%; alert latency > 2.0 s.
Gate G5	GPS Denial Fallback	Disable GPS satellites mid-flight for 30s.	Position drift > 0.5 m/s during total blackout.
Gate G6	End-to-End Mission Victory	Full 4-drone grid search, target alert, and formation RTL.	Mission time > 15 mins; any mid-air collision.

6 Curated Master Resource Library (Free, Paid & Books)

Recommended Books

- **Automate the Boring Stuff with Python** — Al Sweigart (Free to read online)
- **Probabilistic Robotics** — Sebastian Thrun, Wolfram Burgard
- **Computer Networking: A Top-Down Approach** — Kurose & Ross
- **Hands-On Machine Learning with Scikit-Learn, Keras & PyTorch** — Aurélien Géron

Free Video Courses & Official Docs

-  3Blue1Brown Linear Algebra
-  Articulated Robotics ROS 2
-  FreeCodeCamp Sockets
-  Official ROS 2 Jazzy Docs
-  Official PX4 ROS 2 Guide

Paid / Certification Courses (Optional)

- Coursera Deep Learning Specialization (Andrew Ng)
- Udacity Robotics Software Engineer Nanodegree
- edX MIT 6.00.1x CS & Python